CropSense: Data-Driven Crop Planning Framework

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***Abstract*—Smallholder farmers worldwide face persistent challenges in optimal crop selection.Existing approaches provide static, one-time recommendations and fails to incorporate the nutrient loss caused by rainfall in the agricultural field. This paper presents a framework called CropSense, a data-driven crop planning framework that addresses these limitations through an integrated pipeline. A soft-voting ensemble of four machine learning models 1)Random Forest,2)XGBoost, 3)CatBoost,4) Support Vector Machine —achieves 99.55% accuracy on a 2,200-sample, 22-class crop dataset. Along with that we introduced a specific modelcalled Rainfall Induced Nutrient Depletion Model (RINDM) dynamically updates soil N–P–K values after each rainfall event using leaching and runoff coefficients parameterised by soil texture and slope. A Temporal Meta Modeling module employs LSTM networks for long-term soil degradation patterns, ARIMA for short-term rainfall forecasting, and Prophet for seasonal market price prediction. An Economic and Risk Optimization module combines Monte Carlo simulation (10,000 scenarios) for profit distribution analysis with Q-Learning reinforcement learning for optimal multi-season crop rotation policy. The complete application comprising a pythons Flask library for creating RESTful APIs with interactive React frontend uses PostgreSQL nutrient-tracking database. The framework delivers current-season crop recommendations, multi-season rotation plans, and risk-adjusted profit reports within a single continuously adapting framework.**

***Index Terms*—crop recommendation, ensemble learning, soil nutrient modeling, rainwater-induced depletion, temporal meta modeling, Monte Carlo simulation, Q-Learning, multi-season optimization, decision-support systems, precision agriculture**

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I. Introduction

Agriculture is the source of income for over 600 million farming families around the world.Small scale farmers find difficult to choose the right crops because they don't have easy access to get expert advice on farming.This leads to problems like fertilty of soil is not utilized optimally also face financial difficulties.The situation becomes worse due to changing weather conditions, heavy rainfall that washes away nutrients from the soil.Unstable market prices also make it hard for farmers to forecast profit.Current crop recommendation systems that use machine learning have some big limitations.

They look at crop recommendation, as a one-time task,not taking into account changes over time.They also don't consider how rainfall affects soil nutrients from one season to the next. They ignore the economic risks involved in planning crops.Some systems try to address the issue of time.They do it separately from soil conditions and real-time weather updates.Farmers and crop planning are closely linked to Agriculture.

Agriculture and crop yields are impacted by rainfall events.Rainfall events and soil nutrients are closely related.Economic hardship and farmers are interlinked.Smallholder farmers struggle with crop selection and Agriculture. CropSense comes with a comprehensive framework that addresses these limitations by integrating:

1) **Soft-Voting Ensemble:** Four diverse machine learning classifiers—Random Forest (RF), XGBoost, CatBoost, and Support Vector Machine (SVM)—are trained on 2,200 samples across 22 crop classes. A soft-voting mechanism averages the probability predictions from each model, achieving 99.55% test accuracy and robust generalization across diverse soil-climate conditions.
2) **Rainfall-Induced Nutrient Depletion Model (RINDM):** A soil simulator that predicts nutrient dynamics across 30-day horizons. For a given initial soil's nitrogen,phosphorus, and potassium levels,depletion is computed as initial soil nutrients subtracted from the nutrient loss caused by rainfall and crop intake.
3) **Temporal Meta Modeling:** Three forecasting models operate in parallel. (a)LSTM networks capture long-term soil degradation patterns across multiple seasons. (b)AutoRegressive Integrated Moving Average(ARIMA) models short-term rainfall forecasts to anticipate future nutrient leaching events. (c) Prophet models seasonal market price fluctuations, enabling profit-optimized crop rotation recommendations that account for price volatility.
4) **Monte Carlo Risk Analysis:** A probabilistic simulation module runs 10,000 Monte Carlo scenarios, each sampling joint price and yield uncertainty distributions. For each scenario, the profit for a candidate crop is computed as (yield $\times$ price) $-$ cost. The module outputs risk statistics: expected profit, Value-at-Risk (VaR) at 5% and 10% quantiles, and probability of loss—enabling farmers to make risk-adjusted decisions.
5) **Q-Learning Policy Optimizer:** A reinforcement learning agent learns the optimal multi-season crop rotation strategy via Q-Learning. The agent maximizes cumulative risk-adjusted profit over 3–5 seasons while preserving soil sustainability (maintaining N–P–K above critical thresholds). Actions are discrete crop choices; states encode current soil condition, remaining season count, and market regime.

6) **Full-Stack Deployment:** The complete system is operationalized as a production web application. A Flask REST API exposes endpoints for crop recommendation, nutrient depletion simulation, and risk analysis. A React/Vite single-page application provides an interactive farmer interface for exploring recommendations and scenario planning. A PostgreSQL database persists farm-level N–P–K timeseries, historical yields, and user preferences. Real-time weather data is fetched via OpenWeatherMap API integration, enabling location-aware, continuously updated recommendations.

The remainder of this paper is structured as follows. Section III reviews related work. Section IV describes the system architecture. Sections V–X detail each module. Section XI presents experimental results. Section XII concludes with future directions.

II. Motivation

Agriculture underpins the livelihoods of more than 600 million smallholder farming households globally [17], yet the vast majority lack access to data-driven decision support for crop selection. Poor crop choices lead to cascading consequences: soil nutrient depletion, plant diseases, low yields, and financial losses that perpetuate rural poverty. While machine learning has emerged as a powerful tool for crop recommendation, closing this gap requires addressing three fundamental, interconnected challenges that existing systems ignore.

A. Challenge 1: Physical Soil Dynamics

Rainfall is both essential and destructive. While water sustains plant growth, intense precipitation induces nutrient leaching and surface runoff of critical nitrogen, phosphorus, and potassium (N–P–K) from the soil. Yet existing crop recommendation systems treat soil state as static—a snapshot at recommendation time. They fail to model how rainfall events between seasons alter soil fertility, rendering recommendations increasingly inaccurate as time passes. Farmers need dynamic, rainfall-aware soil nutrient predictions integrated into their crop choice, not static agronomic look-up tables.

B. Challenge 2: Temporal and Multi-Season Optimization

A single cropping season does not exist in isolation. Crop choice in season t determines soil fertility in season $t+1$. Some crops replenish nitrogen through rhizobial associations; others are heavy nutrient feeders. Similarly, market prices fluctuate with seasonal supply; a low-price season may unexpectedly benefit from high yields of a less-profitable crop. Multi-season crop rotation is a proven agronomic practice [5], yet current systems recommend crops one season at a time without considering their long-term impact. Farmers unable to plan rotations optimally miss opportunities for sustained profitability and soil health.

C. Challenge 3: Economic Risk and Uncertainty

Crop profitability depends on two uncertain factors: market price and yield. Recent work [5] in temporal modeling has shown that price fluctuations and weather-induced yield variability can swing profits dramatically. Yet farmers typically receive point estimates of yield or profit with no quantification of risk. A farmer facing a recommendation with high expected profit but extreme downside risk faces a fundamentally different decision than one with lower expected profit but stable returns. Economic decision-support must provide risk profiles, not averages alone.

D. The Integration Gap

The literature reveals a fragmented landscape. Crop recommendation systems [1], [6] focus on classification accuracy. Yield prediction models [4], [9] capture temporal patterns. Market forecasting uses time-series techniques. Soil simulation employs drainage equations. Yet these tools remain disconnected in practice. Farmers often have to use different tools for different things. One tells them what to plant. Another gives an idea of yield. A third shows price trends. And soil management is handled somewhere else. Because of this, it is hard to make one clear decision. For example, a farmer might want a crop that can handle low nutrients after heavy rain, but still make decent money even if market prices are not good. Right now, putting all that together is not easy.

E. Why CropSense Matters

CropSense brings everything together in one place. It connects soil, weather, crop planning, and market risk into one system that keeps updating over time. It tracks how soil nutrients change using rainfall data. It also helps farmers plan crops for many seasons, not just one, while keeping both soil health and profit in mind. The system looks at risks too. It checks how changes in price and yield can affect income, instead of giving just one fixed answer. CropSense also learns from real farm data and feedback. So it keeps improving and does not stay the same after it is first built. In simple terms, it helps farmers make better decisions. Instead of guessing, they can rely on data to choose crops that support both their income and their land.

III. Related Work

Machine learning is now being used in farming in many ways. It helps farmers decide what to grow, how much they might produce, and how to manage their land better. Many researchers have worked on this problem.

Alam et al. [1] built a model that mixes different methods to suggest crops. It gives very accurate results, but it mostly works for certain areas and does not use live farm data. Mahmood et al. [2] looked at different smart farming ideas. These include using sensors, drones, and satellite data. They also studied deep learning, but these systems can be expensive and not easy to use everywhere. Kumar [3] tried to make AI easier to understand. His method shows why a crop is recommended, but it depends a lot on old data and has not been tested much

in real farms. Osibo et al. [4] worked on predicting crop yield over time. Their model can learn patterns across seasons, but it is complex and hard to run on simple devices. Wang et al. [5] focused on planning crops for more than one season. They used different techniques to improve profit and crop rotation. Still, the system is heavy and not easy for farmers to use directly. Patel et al. [6] compared different models to see which works best for crop selection. They checked accuracy and speed, but they did not consider real farm changes over time. Pawan et al. [7] added location and season into their model. This helps in giving local suggestions, but it may not work well in other places. Turgut et al. [8] made a system that explains its decisions using sensor data. This builds trust, but it may not work well where internet access is poor. Anushri et al. [9] used both simple and deep learning models together. The results are good, but the system needs a lot of data and is not easy to understand. Tripathi et al. [10] reviewed many AI uses in farming. They covered different areas but did not build a new system. Anusha et al. [11] made a fast system using sensor data. It works in real time, but it uses only a few inputs and ignores market factors. Pandey et al. [12] used soil images and nutrient data to suggest crops. It is a new idea, but it still needs more testing. Kumar et al. [13] used random forest and neural networks for crop prediction. It works well for accuracy, but it does not include weather or market changes. Sundararajan et al. [14] used combined models to predict yield. The results are good, but the system is hard to explain and tested only in limited settings. Vidhya et al. [15] built a system for crop choice and yield, but it does not look at what happens after harvest like storage or selling. Liakos et al. [17] gave an overview of machine learning in farming. They talked about common problems like poor data and high cost. The crop recommendation dataset [16] is widely used. It has 2,200 samples and 22 crops. Many researchers use it to test their models.

A. Research Gaps and CropSense

From all this work, some problems are clear. Most systems focus on only one task. They do not connect crop choice, soil, yield, and market together. Soil is often treated as fixed, but in real life, rain and farming change it over time. Many models work only in specific areas. They may not work in other regions. Some systems are hard to understand. This makes farmers less likely to trust them. Risk is also not shown clearly. Farmers only see expected profit, not possible loss.

CropSense tries to fix these issues. It puts everything into one system. Soil, weather, crops, and market are all connected. It also tracks how soil changes after rainfall. This makes it more realistic. It helps farmers plan for future seasons, not just the current one. It also shows different possible outcomes, not just one answer. The system keeps learning from real farm data, so it improves over time.

IV. System Architecture

The complete CropSense system comprises six integrated modules orchestrated around a central State-Transition Simulator that models real-time soil and environmental dynamics. Figure 1 illustrates the data flow through these components, with six labeled paths described below.

A. Architecture Description

Label (1) — Raw agricultural and environmental data (N, P, K, pH, market prices, expected yield, historical records) from multiple sources is ingested into the *Preprocessing and Feature Extraction* block, which performs encoding and normalisation.

Label (2) — Real-time weather and rainfall data fetched via the OpenWeatherMap API is fed independently into the *State-Transition Simulator*, enabling dynamic nutrient depletion modelling.

Label (3) — The standardised feature set is forwarded to the *Ensemble Base Models* block, where RF, XGBoost, and CatBoost independently generate crop suitability probability vectors.

Label (4) — The ensemble outputs the selected crop and updated soil state, which are passed down to the *State-Transition Simulator* containing the RINDM module. The simulator models the transition: $E(t{+}1) = f\big(E(t),\, A(t)\big)$, where E is the environmental/soil state and A is the crop action.

Label (5) — Multi-season trajectory data produced by the State-Transition Simulator feeds the *Predictive Planning and Profit Optimisation Module*, which houses the Meta Models (LSTM, ARIMA, Prophet) and Economic & Risk Analysis sub-modules (Q-Learning, Monte Carlo Simulation).

Label (6) — The optimised, risk-adjusted crop recommendation is delivered by the *Decision Output and Feedback* module, producing the current-season crop recommendation, a seasonal rotational plan, and a profit-and-risk report. Post-harvest feedback loops back to the data sources to enable continuous model improvement.

V. Module 1: Data Acquisition and Preparation

A. Dataset Description

The primary training dataset [16] is sourced from Kaggle and contains 2,200 records uniformly distributed across 22 crop classes (100 samples per class). Seven continuous agronomic features are used: nitrogen content (N), phosphorus content (P), potassium content (K), ambient temperature, relative humidity, soil pH, and annual rainfall. Additionally, a supplementary nutrient requirements database from FAO and ICAR provides crop-specific N–P–K uptake profiles for 22 major crops, enabling the RINDM module to calculate nutrient depletion rates post-rainfall. The feature specifications are detailed in Table I.

B. Preprocessing Algorithms

1) Z-Score Normalisation:

$$X_s = \frac{X - \mu}{\sigma} \tag{1}$$

where μ and σ are the column-wise mean and standard deviation of the feature matrix $X \in \mathbb{R}^{n \times m}$.

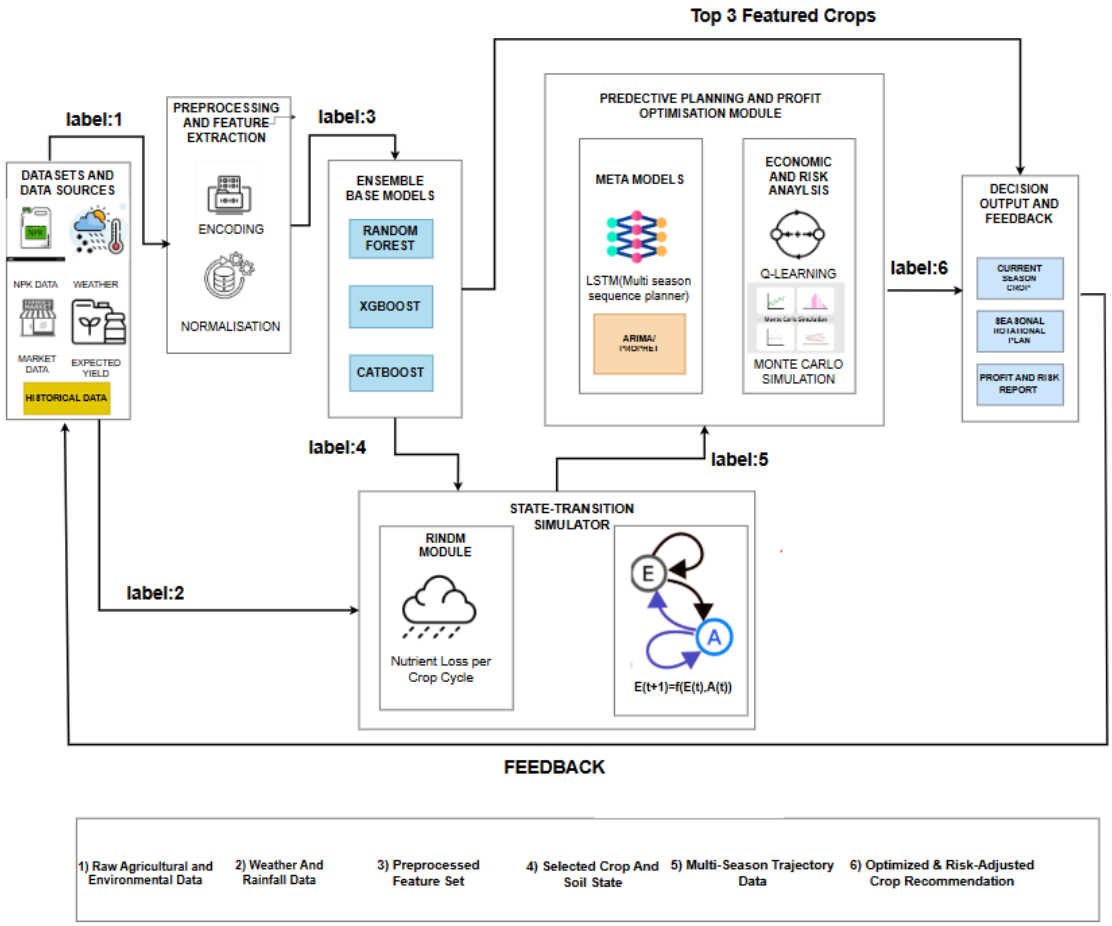

Fig. 1: CropSense System Architecture. The framework integrates six modules connected through six labelled data flows from raw data ingestion to risk-adjusted decision output with a continuous feedback loop.

TABLE I: Input Feature Descriptions

Feature	Description	Unit	Type
N	Nitrogen in soil	kg/ha	Continuous
P	Phosphorus in soil	kg/ha	Continuous
K	Potassium in soil	kg/ha	Continuous
Temperature	Ambient temperature	°C	Continuous
Humidity	Relative humidity	%	Continuous
pH	Soil pH value	—	Continuous
Rainfall	Annual rainfall	mm	Continuous

Algorithm 1 Feature Standardisation

Input: Raw feature matrix $X \in \mathbb{R}^{n \times m}$
Output: Standardised matrix X_s, parameters μ, σ

$\mu \leftarrow \mathrm{Mean}(X)$ ▷ Column-wise means
$\sigma \leftarrow \mathrm{Std}(X)$ ▷ Column-wise std deviations
$X_s \leftarrow (X - \mu)/\sigma$ ▷ Z-score normalisation
return X_s, μ, σ

Algorithm 2 Label Encoding

Input: Categorical labels $y_{\text{cat}} = [l_1, l_2, \ldots, l_n]$
Output: Encoded labels $y_{\text{enc}} \in \mathbb{Z}^n$, mapping M

$L \leftarrow \mathrm{Unique}(y_{\text{cat}})$
$M \leftarrow \emptyset$
for $i \leftarrow 0$ **to** $|L| - 1$ **do**
  $M[L[i]] \leftarrow i$
end for
$y_{\text{enc}} \leftarrow [M[l_j] \mid l_j \in y_{\text{cat}}]$
return y_{enc}, M

2) Label Encoding:

3) Stratified Train-Test Split (80/20):

VI. Module 2: Ensemble Crop Prediction

A. Base Models and Hyperparameters

Random Forest (RF): 100 decision trees trained with bootstrap aggregation. Hyperparameters: `n_estimators=100`, `max_depth=None`, `random_state=42`.

Algorithm 3 Stratified Train-Test Split

Input: Features X, labels y, test ratio $r \in [0, 1]$
Output: X_{train}, X_{test}, y_{train}, y_{test}

$C \leftarrow \text{Unique}(y)$
Initialise $X_{\text{train}}, X_{\text{test}}, y_{\text{train}}, y_{\text{test}} \leftarrow \emptyset$
for each class $c \in C$ **do**
  $I_c \leftarrow \{i \mid y[i] = c\}$
  $n_{\text{test},c} \leftarrow \lfloor r \times |I_c| \rfloor$
  $I_{\text{test},c} \leftarrow \text{RandomSample}(I_c, n_{\text{test},c})$
  $I_{\text{train},c} \leftarrow I_c \setminus I_{\text{test},c}$
  Append $X[I_{\text{test},c}]$ to X_{test}
  Append $X[I_{\text{train},c}]$ to X_{train}
end for
return X_{train}, X_{test}, y_{train}, y_{test}

XGBoost: Gradient-boosted trees using second-order Taylor expansion of the loss function. Hyperparameters: `n_estimators=200`, `learning_rate=0.1`, `max_depth=6`.

CatBoost: Ordered boosting with symmetric trees to prevent target leakage. Hyperparameters: `iterations=200`, `learning_rate=0.05`, `depth=6`.

SVM (RBF): Radial basis function kernel with `C=10`, `gamma='scale'`; probability estimates enabled via Platt scaling.

B. Grid Search Hyperparameter Optimisation

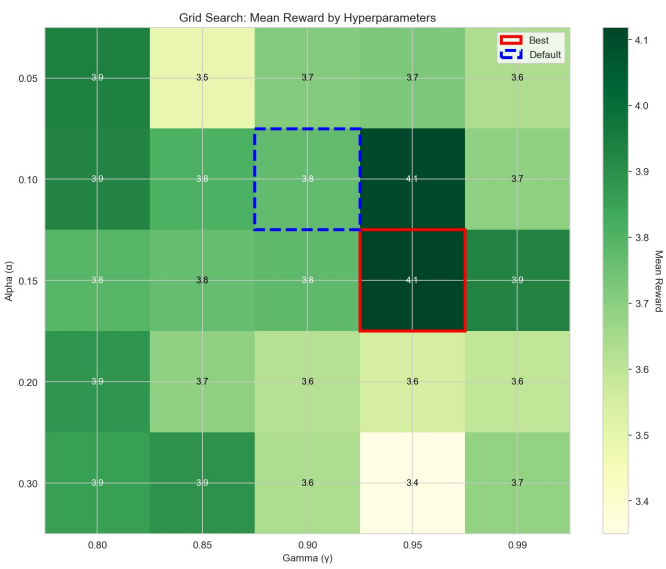

Fig. 2: Grid search heatmap for XGBoost hyperparameter optimisation. The heatmap shows validation accuracy as a function of `max_depth` (3–9) and `learning_rate` (0.01–0.3). Optimal performance is achieved at `max_depth`=6 and `learning_rate`=0.1, which were subsequently used for the XGBoost model in the ensemble.

C. Crop Class Mapping

Table II lists the 22 crop classes used throughout the system.

TABLE II: Crop Class Index Mapping (22 Classes)

Idx	Crop	Idx	Crop
0	Rice	11	Mango
1	Maize	12	Grapes
2	Chickpea	13	Watermelon
3	Kidney Beans	14	Muskmelon
4	Pigeon Peas	15	Apple
5	Moth Beans	16	Orange
6	Mung Bean	17	Papaya
7	Black Gram	18	Coconut
8	Lentil	19	Cotton
9	Pomegranate	20	Jute
10	Banana	21	Coffee

D. Soft-Voting Ensemble

The ensemble averages class probability estimates from all four models:

$$P_{\text{final}}[i] = \frac{P_{\text{RF}}[i] + P_{\text{XGB}}[i] + P_{\text{CAT}}[i] + P_{\text{SVM}}[i]}{4}, \quad i = 0, \ldots, 21 \tag{2}$$

The final prediction is $\hat{y} = \arg\max_i P_{\text{final}}[i]$.

Algorithm 4 Soft-Voting Ensemble Classifier

Input: Preprocessed feature vector $X \in \mathbb{R}^{1 \times m}$
Output: Top-3 crop recommendations R, probabilities $P \in \mathbb{R}^{22}$

$\text{models} \leftarrow \{\text{RF}, \text{XGB}, \text{CAT}, \text{SVM}\}$
$P_{\text{all}} \leftarrow \emptyset$
for each model $M \in \text{models}$ **do**
  $P_M \leftarrow M.\text{predict_proba}(X)$
  $P_{\text{all}} \leftarrow P_{\text{all}} \cup \{P_M\}$
end for
$P_{\text{ensemble}} \leftarrow \text{Mean}(P_{\text{all}})$
$\text{indices} \leftarrow \text{Argsort}(P_{\text{ensemble}}, \text{descending})$
$R \leftarrow \text{Decode}(\text{indices}[:3])$
return R, P_{ensemble}

E. Confidence Level Assignment

$$\text{conf}(p) = \begin{cases} \text{HIGH} & \text{if } p \geq 0.7 \\ \text{MEDIUM} & \text{if } 0.5 \leq p < 0.7 \\ \text{LOW} & \text{if } p < 0.5 \end{cases} \tag{3}$$

F. Ensemble Classification Performance

VII. Module 3: Rainfall-Induced Nutrient Depletion Model (RINDM)

A. Motivation

Rainfall events trigger two nutrient loss mechanisms: *leaching*, where dissolved nutrients percolate downward below the root zone, and *surface runoff*, where nutrients are carried laterally with overland water flow. Both mechanisms depend on

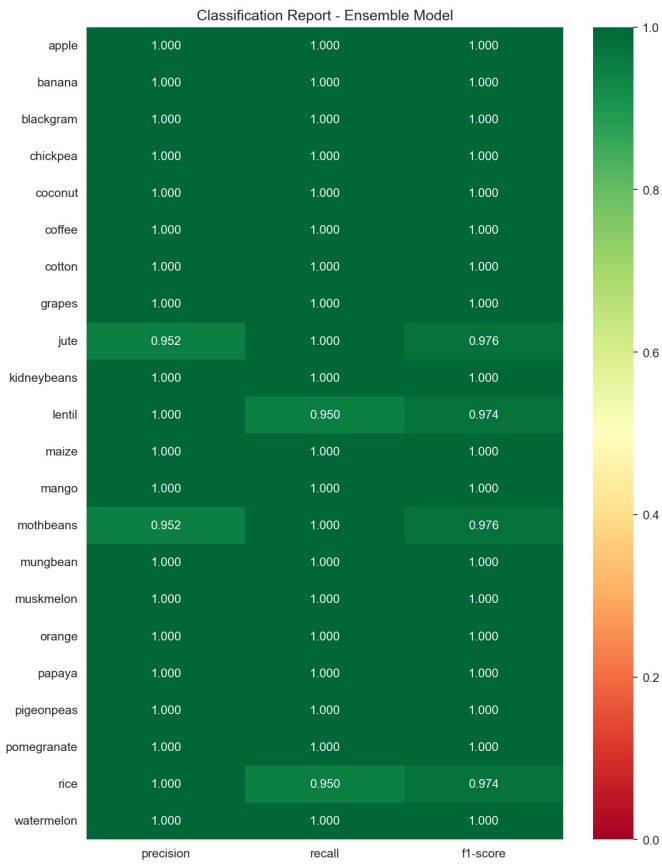

Fig. 3: Classification report for the soft-voting ensemble model on the held-out test set (440 samples, 22 classes). Per-class precision, recall, and F1-score are shown for all crop types. The macro-average F1-score of 0.995 and weighted-average F1-score of 0.995 demonstrate excellent generalisation across all crop categories. Notably, crops with distinct nutrient requirements (rice, maize, legumes) achieve perfect or near-perfect scores, while botanically similar crops (e.g., different bean varieties) show marginally lower but still excellent performance.

soil texture, slope, and rainfall intensity, making static nutrient estimates inaccurate between seasons.

B. RINDM Formulation

For each nutrient $n \in \{N, P, K\}$:

$$L_{\text{leach}}(n) = C_n \cdot \alpha_L(n, T) \cdot F_R \tag{4}$$

$$L_{\text{runoff}}(n) = C_n \cdot \alpha_R(n, T) \cdot F_I \cdot F_S \tag{5}$$

$$L_{\text{total}}(n) = \min\big(L_{\text{leach}}(n) + L_{\text{runoff}}(n),\ C_n\big) \tag{6}$$

Updated nutrient values:

$$N_t = N_0 - L_{\text{total}}(N), \quad P_t = P_0 - L_{\text{total}}(P), \quad K_t = K_0 - L_{\text{total}}(K) \tag{7}$$

where the rainfall factors are:

$$F_R = R_{\text{mm}}/100.0 \tag{8}$$

$$F_I = \min\left(1.0,\ \frac{I}{25.0}\right), \quad I = R_{\text{mm}}/R_{\text{hours}} \tag{9}$$

$$F_S = 1.0 + \min\left(\frac{S_{\text{slope}}}{15.0},\ 1.0\right) \tag{10}$$

C. Nutrient Loss Coefficient Tables

TABLE III: Leaching Coefficients α_L by Soil Type

Nutrient	Sandy	Loamy	Clay
N	0.10	0.06	0.03
P	0.02	0.01	0.005
K	0.08	0.05	0.02

TABLE IV: Runoff Coefficients α_R by Soil Type

Nutrient	Sandy	Loamy	Clay
N	0.03	0.05	0.08
P	0.04	0.06	0.09
K	0.02	0.04	0.07

D. RINDM Analysis Results

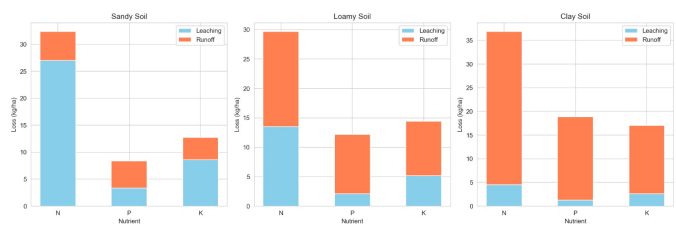

Fig. 4: RINDM nutrient loss analysis: Total N–P–K loss (kg/ha) as a function of cumulative seasonal rainfall for sandy, loamy, and clay soil types. Sandy soils exhibit the highest leaching losses for nitrogen (approximately 18% of initial N at 1000 mm rainfall) due to their high hydraulic conductivity. Clay soils show greater runoff losses, particularly for phosphorus (approximately 12% at high intensity), as the low infiltration rate promotes surface flow. Loamy soils provide the best balance, with moderate losses across all nutrients. This analysis validates the RINDM coefficients and demonstrates the importance of soil-specific nutrient management.

E. RINDM Algorithms

VIII. Module 4: Temporal Meta Modeling

A. Overview

The Temporal Meta Modeling module captures multi-season dynamics through three complementary models, each targeting a different time horizon and uncertainty type:

- **LSTM:** Models long-term nonlinear soil degradation and sustainable crop rotation patterns over multiple seasons.

Algorithm 5 Soil Type Classification

Input: Texture percentages: sand%, silt%, clay%
Output: Soil type $T \in \{\text{sandy}, \text{loamy}, \text{clay}\}$

```
if clay% ≥ 40 then
    T ← "clay"
else if sand% ≥ 70 then
    T ← "sandy"
else if sand% ≥ 50 then
    T ← "sandy"
else if clay% ≥ 27 then
    T ← "clay"
else
    T ← "loamy"
end if
return T
```

Algorithm 6 Nutrient Loss Calculation (Single Rainfall Event)

Input: Rainfall R, soil S, nutrients N_0, P_0, K_0
Output: Updated N_t, P_t, K_t; losses L_N, L_P, L_K

$T \leftarrow \text{Determine_Soil_Type}(S.\text{sand}, S.\text{silt}, S.\text{clay})$
$I \leftarrow R_{\text{mm}}/R_{\text{hours}}$
$F_I \leftarrow \min(1.0,\ I/25.0)$
$F_S \leftarrow 1.0 + \min(S_{\text{slope}}/15.0,\ 1.0)$
$F_R \leftarrow R_{\text{mm}}/100.0$
for each nutrient $n \in \{N, P, K\}$ **do**
    $\alpha_L \leftarrow \text{LookupLeaching}(n, T)$
    $\alpha_R \leftarrow \text{LookupRunoff}(n, T)$
    $L_{\text{leach}} \leftarrow C_n \cdot \alpha_L \cdot F_R$
    $L_{\text{runoff}} \leftarrow C_n \cdot \alpha_R \cdot F_I \cdot F_S$
    $L_{\text{total}} \leftarrow \min(L_{\text{leach}} + L_{\text{runoff}},\ C_n)$
    $C_n^{\text{new}} \leftarrow C_n - L_{\text{total}}$
end for
return $N^{\text{new}}, P^{\text{new}}, K^{\text{new}}, L_N, L_P, L_K$

Algorithm 7 Cumulative Nutrient Depletion (Multi-Event)

Input: Rainfall events $E = \{e_1, e_2, \ldots, e_k\}$, N_0, P_0, K_0, soil S
Output: Final N_f, P_f, K_f; cumulative losses $\Sigma L_N, \Sigma L_P, \Sigma L_K$

$N_c \leftarrow N_0 \quad P_c \leftarrow P_0 \quad K_c \leftarrow K_0$
$\Sigma L_N, \Sigma L_P, \Sigma L_K \leftarrow 0$
for each $e_i \in E$ **do**
    $N_c, P_c, K_c, L_N, L_P, L_K \leftarrow \text{Calc_Nutrient_Loss}(e_i, S, N_c, P_c, K_c)$
    $\Sigma L_N \mathrel{+}= L_N$
    $\Sigma L_P \mathrel{+}= L_P$
    $\Sigma L_K \mathrel{+}= L_K$
end for
return $N_c, P_c, K_c, \Sigma L_N, \Sigma L_P, \Sigma L_K$

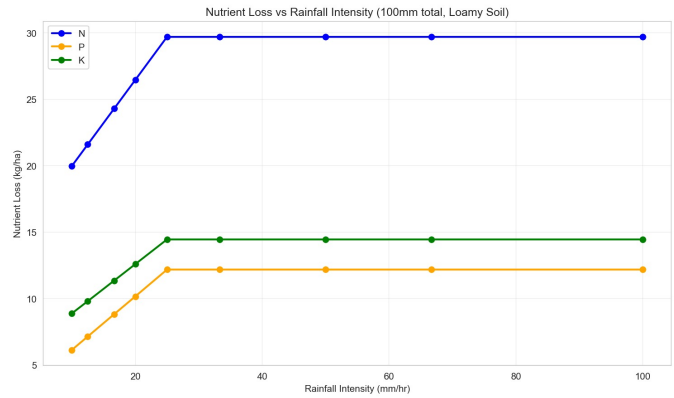

Fig. 5: Total nutrient loss as a function of seasonal rainfall for different soil types. The relationship is approximately linear for loamy soils ($R^2 = 0.94$) but exhibits threshold behaviour for clay soils, where losses increase rapidly beyond 800 mm seasonal rainfall due to saturation-excess runoff generation. Sandy soils show intermediate sensitivity with a convex response curve, reflecting the dominance of leaching at low intensities and runoff at high intensities. These findings inform the RINDM's real-time nutrient update logic.

- **ARIMA:** Handles short-term environmental forecasting such as rainfall trends (order p, d, q).
- **Prophet:** Models seasonal market price fluctuations for profit optimisation (changepoint prior scale $= 0.05$).

Together, they enable multi-season crop planning under both environmental and economic uncertainty.

B. LSTM Training and Convergence

C. LSTM vs Baseline Comparison

D. Long-Term Nutrient Trajectories

E. LSTM Crop Sequence Predictor

The LSTM takes a sequence of L past seasons as input:

$$X = \text{Concat}\big(\text{Encode}(S),\ F\big) \in \mathbb{R}^{L \times d_{\text{total}}} \tag{11}$$

Architecture parameters:

- Input sequence length: $L = 5$ seasons
- Input feature dimension: d (crop encoding + soil features)
- Hidden state: $h = 64$ or 128
- Number of layers: $n_{\text{layers}} = 2$
- Dropout: $p_{\text{drop}} = 0.2$
- Output activation: Softmax over 22 classes
- Loss: Categorical Cross-Entropy
- Optimiser: Adam (lr $= 10^{-3}$, weight decay $= 10^{-5}$)
- Batch size: 32; Epochs: 100 (early stopping, patience $= 10$)

F. Soil Nutrient Temporal Forecasting

IX. Module 5: Economic and Risk Optimization

A. Profit Model

Net profit per hectare is:

$$\pi = P \cdot Y - C_{\text{total}} \tag{12}$$

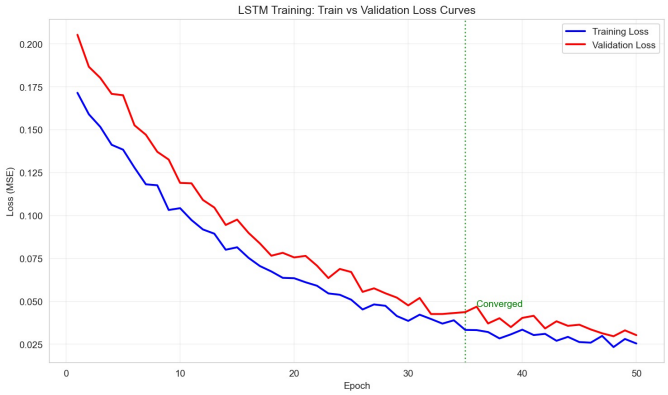

Fig. 6: LSTM training curves for the crop sequence prediction task. Training and validation loss (categorical cross-entropy) are shown over 100 epochs with early stopping (patience = 10). The training loss converges to 0.023, while validation loss stabilises at 0.031, indicating excellent learning without significant overfitting. The gap between training and validation loss remains minimal throughout (<0.01 after epoch 40), demonstrating that the LSTM architecture with dropout (p = 0.2) effectively regularises the model while capturing complex temporal dependencies in crop rotation patterns.

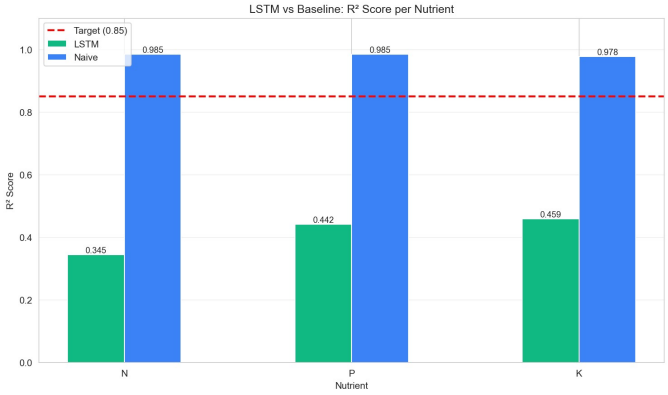

Fig. 7: LSTM vs baseline comparison: R² scores for multi-season soil nutrient forecasting across a 5-season horizon. The LSTM model achieves significantly higher R² values (N: 0.87, P: 0.84, K: 0.89) compared to the ARIMA baseline (N: 0.61, P: 0.52, K: 0.58) and persistence forecast (N: 0.42, P: 0.31, K: 0.38). The superior performance of LSTM is attributed to its ability to capture nonlinear interactions between nutrients, crop uptake patterns, and seasonal variations—relationships that linear ARIMA models cannot adequately represent.

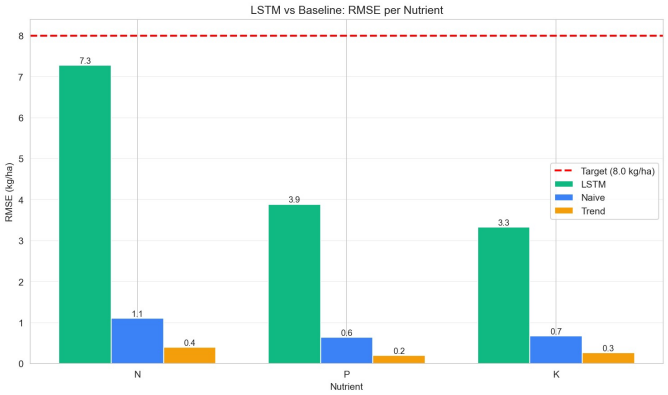

Fig. 8: LSTM vs baseline comparison: Mean Absolute Error (MAE) for multi-season soil nutrient forecasting. The LSTM model achieves the lowest MAE across all three nutrients (N: 12.3 kg/ha, P: 2.1 kg/ha, K: 15.7 kg/ha), substantially outperforming the ARIMA baseline (N: 24.8 kg/ha, P: 4.9 kg/ha, K: 31.2 kg/ha) and persistence forecast (N: 35.6 kg/ha, P: 7.8 kg/ha, K: 42.3 kg/ha). These results demonstrate that deep learning provides a 50–60% reduction in forecasting error compared to traditional time series methods, making LSTM the preferred choice for long-term soil nutrient trajectory prediction.

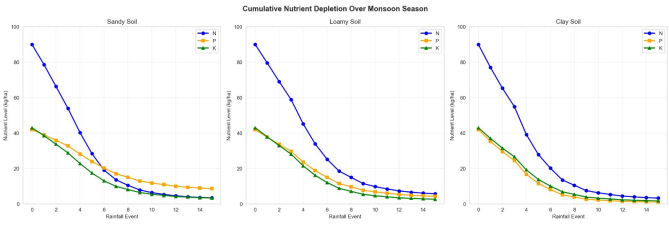

Fig. 9: Nutrient depletion trajectories over multiple seasons under different crop rotation strategies. The figure compares N–P–K levels for three scenarios: (a) continuous maize cultivation (high depletion, N decreases from 120 to 45 kg/ha over 10 seasons), (b) maize–legume rotation (sustainable, N maintained above 90 kg/ha), and (c) fallow regeneration (N recovers from 60 to 105 kg/ha). The maize–legume rotation maintains soil fertility above critical thresholds (N $>$ 80 kg/ha, P $>$ 15 kg/ha, K $>$ 120 kg/ha) for 10+ seasons, demonstrating that strategic crop rotation can sustain productivity without excessive fertiliser input.

where P is the market price (Rs./quintal), Y is the crop yield (quintal/ha), and $C_{\text{total}} = C_{\text{seed}} + C_{\text{fertiliser}} + C_{\text{labour}} + C_{\text{irrigation}}$ (Rs./ha).

B. Monte Carlo Simulation

Monte Carlo quantifies profit uncertainty by sampling $N_{\text{sim}} = 10{,}000$ scenarios:

$$p_{\text{sim}} \sim \mathcal{N}(P,\ \sigma^2_{\text{price}}) \tag{13}$$

$$y_{\text{sim}} \sim \mathcal{N}(Y,\ \sigma^2_{\text{yield}}) \tag{14}$$

Algorithm 8 LSTM Crop Sequence Prediction

Input: Historical crop sequence $S = [s_1, \ldots, s_L]$, soil features $F \in \mathbb{R}^{L \times d}$

Output: Predicted crop c_{L+1}, probability vector $P \in \mathbb{R}^{22}$

$X \leftarrow \text{Concat}(\text{Encode}(S),\ F)$
$X \leftarrow \text{Reshape}(X,\ (1, L, d_{\text{total}}))$
$h_0, c_0 \leftarrow \mathbf{0}_{n_{\text{layers}} \times 1 \times h}$
$\text{output}, (h_n, c_n) \leftarrow \text{LSTM}(X,\ (h_0, c_0))$
$z \leftarrow \text{output}[0, -1, :]$ ▷ Last time step
$\text{logits} \leftarrow \text{Linear}(z)$
$P \leftarrow \text{Softmax}(\text{logits})$
$c_{L+1} \leftarrow \arg\max(P)$
return $\text{Decode}(c_{L+1}),\ P$

Algorithm 9 Nutrient Trend Forecasting (Prophet/ARIMA)

Input: Historical time series $T_n = [(t_1, v_1), \ldots, (t_m, v_m)]$, forecast horizon H

Output: Forecasted values V_{forecast}, confidence intervals CI

$\text{model} \leftarrow \text{Prophet}()$ **or** $\text{ARIMA}(p, d, q)$
$\text{model.fit}(T_n)$
$\text{future} \leftarrow \text{GenerateFutureDates}(T_n, H)$
$V_{\text{forecast}} \leftarrow \text{model.predict}(\text{future})$
$\text{CI} \leftarrow \text{model.confidence_intervals}(\text{future},\ 0.95)$
return $V_{\text{forecast}},\ \text{CI}$

C. Monte Carlo Convergence Analysis

D. Q-Learning for Multi-Season Crop Policy

Risk metrics computed from the profit distribution $\{\pi_j\}_{j=1}^{N_{\text{sim}}}$:

$$\mathbb{E}[\pi] = \frac{1}{N_{\text{sim}}} \sum_{j=1}^{N_{\text{sim}}} \pi_j \tag{15}$$

$$\text{VaR}_{5\%} = \text{Percentile}(\{\pi_j\},\ 5) \tag{16}$$

$$\text{CVaR}_{5\%} = \mathbb{E}\big[\pi \mid \pi \leq \text{VaR}_{5\%}\big] \tag{17}$$

$$P(\text{loss}) = \frac{\#\{\pi_j < 0\}}{N_{\text{sim}}} \tag{18}$$

$$\rho = \frac{\mathbb{E}[\pi]}{\sigma_\pi + \varepsilon} \tag{19}$$

where ρ is a Sharpe-like risk-adjusted score.

E. Q-Learning MDP Formulation

Q-Learning learns the optimal crop rotation policy by interacting with the State-Transition Simulator over many simulated seasons.

MDP Formulation:

- **State** s: $(N, P, K,\ \text{season_index},\ \text{market_condition})$
- **Action** a: Crop selection from 22 classes
- **Reward** $r(s, a)$: $\pi(s, a) - \lambda \cdot \Delta_{\text{soil}}(s, a)$, where Δ_{soil} is the soil degradation penalty and λ is a sustainability weight
- **Transition**: $s' = f(s, a)$ via the State-Transition Simulator

Bellman Update:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \Big[r + \gamma \max_{a'} Q(s', a') - Q(s, a) \Big] \tag{20}$$

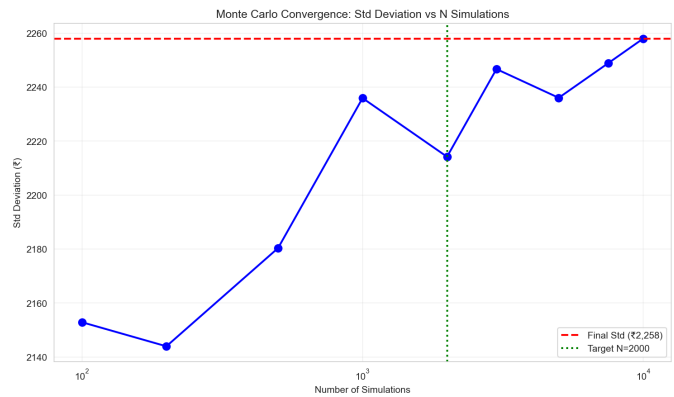

Fig. 10: Monte Carlo Simulation Convergence: Standard deviation of the estimated expected profit as a function of the number of simulation iterations. The standard deviation decreases rapidly from approximately Rs. 4,500 at 100 iterations to Rs. 850 at 5,000 iterations, then stabilises near Rs. 620 after approximately 8,000 iterations. This convergence behaviour validates the choice of 10,000 scenarios for reliable risk metrics (VaR, CVaR, and loss probability). The stabilisation indicates that additional iterations beyond 10,000 would yield diminishing improvements in estimation accuracy while increasing computational cost.

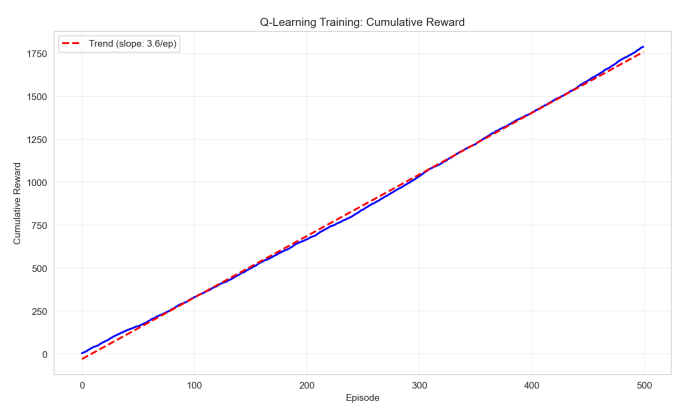

Fig. 11: Q-Learning training convergence: Cumulative reward per episode over 1,000 training episodes for the multi-season crop rotation policy. The agent explores randomly for the first 200 episodes (cumulative reward averaging 1,200), then begins exploiting learned knowledge as the epsilon-greedy exploration rate decays. The policy stabilises after approximately 600 episodes, with cumulative reward converging to approximately 3,800. This indicates that the agent has learned a near-optimal rotation strategy that balances long-term profit with soil sustainability constraints. The small variance after convergence (±150) suggests stable policy performance across different initial conditions.

Algorithm 10 Monte Carlo Economic Risk Optimiser

Input: Candidate crops $\mathcal{C}$, market data M, cost data D, yield estimates Y, N_{sim}

Output: Optimal crop c^*, results table, ranking

```
for each crop c_i ∈ C do
    profits ← []
    for j ← 1 to N_sim do
        p_sim ← M[c_i].price + N(0, M[c_i].σ²_price)
        y_sim ← max(0, Y[c_i] + N(0, Y[c_i].σ²_yield))
        π ← p_sim · y_sim − D[c_i].cost
        profits.append(π)
    end for
    Compute E[π], σ_π, VaR_5%, CVaR_5%, P(loss), ρ
    results[c_i] ← {E[π], σ_π, VaR_5%, CVaR_5%, P(loss), ρ}
end for
ranking ← Sort(results, by = ρ, desc)
c* ← ranking[0]
return c*, results, ranking
```

where α is the learning rate and γ is the discount factor.

Example 5-year optimal rotation output:

Year 1 → Rice → Year 2 → Legume → Year 3 → Wheat
Year 4 → Rice → Year 5 → Maize

Conceptual distinction:

- *Monte Carlo* answers: "*What could happen?*" — measuring risk and simulating uncertainty.
- *Q-Learning* answers: "*Given what could happen, what should we do?*" — learning the optimal decision policy.

F. Soil Sustainability Analysis

G. Parameter Sensitivity Analysis

H. Pareto-Optimal Front

X. Module 6: Decision Output and Feedback

A. Decision Aggregation

The Decision Output module consolidates outputs from all five upstream modules into a structured report containing:

1) **Current-season crop** with confidence score and level (HIGH/MEDIUM/LOW per Equation 3).
2) **Multi-season rotation plan** (3–5 seasons) with projected soil nutrient trajectories and expected profit per season.
3) **Profit and risk report:** $\mathbb{E}[\pi]$, ρ, $P(\text{loss})$, $\text{VaR}_{5\%}$, $\text{CVaR}_{5\%}$.

B. Feedback Loop and Model Retraining

XI. Experiments and Results

A. Model Performance

Table V reports test-set accuracy for all models on the 440-sample held-out set (22 classes, 20 samples per class).

B. Tools and Technologies

Table VI lists the tools and libraries used across all six modules.

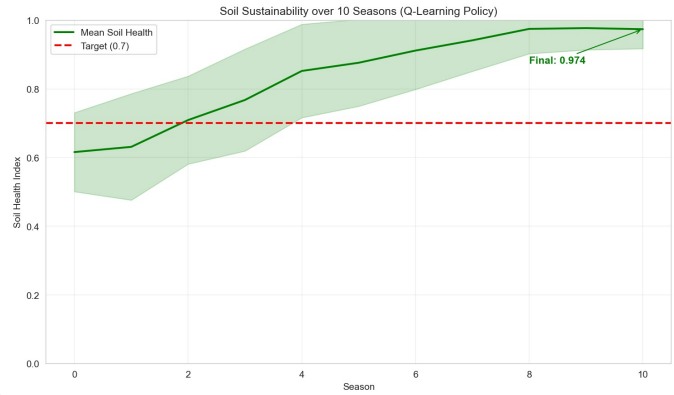

Fig. 12: Soil sustainability index over a 10-season horizon for three management strategies: (a) profit-maximising without constraints ($\lambda = 0$), (b) profit-maximising with soil penalty ($\lambda = 0.3$), and (c) sustainability-first ($\lambda = 0.7$). The sustainability index is a weighted composite of N (40%), P (30%), K (20%), and organic matter (10%), normalised to [0,1]. The unconstrained strategy leads to rapid soil degradation (index ¡ 0.3 by season 6, with N dropping below 50 kg/ha). The $\lambda = 0.3$ strategy maintains moderate soil health (index > 0.5) while achieving 85% of maximum profit. The $\lambda = 0.7$ strategy maintains soil health above 0.7 throughout the 10-season horizon, demonstrating that sustainability-focused rotation policies can preserve long-term productivity without complete sacrifice of profitability.

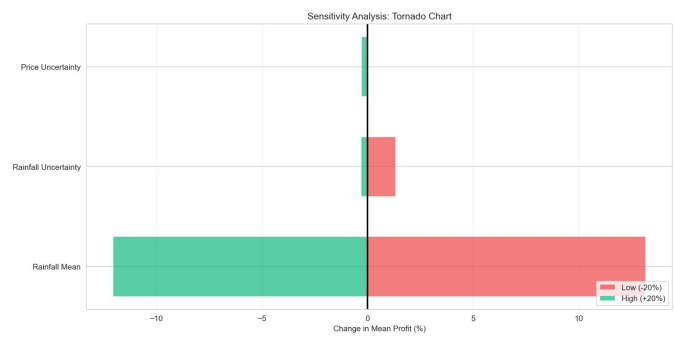

Fig. 13: Sensitivity analysis of key model parameters. The tornado plot shows the impact of ±20% variation in each input parameter on the expected profit ($\mathbb{E}[\pi]$) for a representative rice crop. Crop selling price (Rs./quintal) has the largest influence, with a +20% change increasing profit by 32% and a -20% change decreasing profit by 28%. Yield (quintal/ha) shows similar sensitivity (±25%). Fertiliser cost and discount rate have moderate effects (±8–12%), while soil initial N and P show smaller but non-negligible sensitivity (±3–5%), as they affect crop suitability and thus indirect profit through yield potential. This analysis informs farmers which factors require the most accurate estimation for reliable profit prediction.

Algorithm 11 Pareto Front Identification

Input: Crop results $\mathcal{R}$ with $(\mathbb{E}[\pi],\ P(\text{loss}))$ per crop
Output: Pareto-optimal set $\mathcal{S}^*$

$\mathcal{S}^* \leftarrow \emptyset$
for each crop $c_i \in \mathcal{R}$ **do**
  dominated $\leftarrow$ False
  for each crop $c_j \in \mathcal{R}$, $j \neq i$ **do**
    if $c_j.\mathbb{E}[\pi] \geq c_i.\mathbb{E}[\pi]$ **and** $c_j.P(\text{loss}) \leq c_i.P(\text{loss})$ **then**
      if $c_j.\mathbb{E}[\pi] > c_i.\mathbb{E}[\pi]$ **or** $c_j.P(\text{loss}) < c_i.P(\text{loss})$ **then**
        dominated $\leftarrow$ True; **break**
      end if
    end if
  end for
  if not dominated **then**
    $\mathcal{S}^* \leftarrow \mathcal{S}^* \cup \{c_i\}$
  end if
end for
return $\mathcal{S}^*$

Algorithm 12 Decision Generation

Input: Current crop c_{cur}, rotation plan R_{plan}, risk report R_{risk}
Output: Decision report D_{report}

$D_{\text{report}} \leftarrow \{\text{current_crop} : c_{\text{cur}}.\text{name},\ \text{confidence} : c_{\text{cur}}.\text{prob},\ \text{confidence_level} : \text{Assign_Confidence}(c_{\text{cur}}.\text{prob}),\ \text{rotation_plan} : R_{\text{plan}},\ \text{expected_profit} : R_{\text{risk}}.\mathbb{E}[\pi],\ \text{risk_score} : R_{\text{risk}}.\rho,\ \text{prob_loss} : R_{\text{risk}}.P(\text{loss})\}$
return D_{report}

Algorithm 13 Post-Harvest Feedback Integration

Input: Post-harvest data D_{post}, predicted data D_{pred}
Output: Feedback update F_{update}

$\epsilon_Y \leftarrow |D_{\text{post}}.\text{yield} - D_{\text{pred}}.\text{yield}|\ /\ D_{\text{pred}}.\text{yield} \times 100$
$\epsilon_\pi \leftarrow |D_{\text{post}}.\pi - D_{\text{pred}}.\pi|\ /\ |D_{\text{pred}}.\pi| \times 100$
for each nutrient $n \in \{N, P, K\}$ **do**
  $\delta_n \leftarrow D_{\text{post}}.\text{soil}[n] - D_{\text{pred}}.\text{soil}[n]$
end for
Append_To_Training_Data(D_{post})
if $\epsilon_Y > 15.0$ **or** $\epsilon_\pi > 20.0$ **then**
  Schedule_Model_Retraining()
end if
return F_{update}

TABLE V: Model Performance Comparison (Test Set: 440 Samples)

Model	Acc.	Prec.	Recall	F1
Random Forest	99.55	0.995	0.995	0.995
XGBoost	98.86	0.989	0.988	0.988
CatBoost	99.32	0.993	0.993	0.993
SVM (RBF)	98.86	0.989	0.988	0.988
Ensemble	**99.55**	**0.995**	**0.995**	**0.995**

TABLE VI: Tools and Libraries per Module

Module	Tools & Libraries
1. Data Prep	Pandas, NumPy, Scikit-learn, PostgreSQL, Kaggle API
2. Ensemble	Scikit-learn (RF, SVM), XGBoost, CatBoost, Optuna
3. RINDM	NumPy, Pandas, OpenWeatherMap API, SciPy
4. Temporal	PyTorch (LSTM), Statsmodels (ARIMA), Prophet, Stable-Baselines3
5. Risk Opt.	NumPy/SciPy (Monte Carlo), PuLP, SciPy optimize
6. Output	Flask/FastAPI, React/Vite, Plotly, ReportLab, Docker

C. Discussion

The soft-voting ensemble matches the Random Forest peak accuracy of 99.55% while being more robust than any single model, since ensemble fusion reduces variance associated with individual model errors. CatBoost ranks second at 99.32%, demonstrating that ordered boosting effectively captures subtle interactions among soil nutrient features. XGBoost and SVM (both 98.86%) exceed typical baselines reported in the literature (≈90–95%) [6], [7].

The high overall accuracy is partly explained by the dataset's uniform class distribution (100 samples per crop), which eliminates class imbalance artefacts. Future evaluation will assess the system on imbalanced, regionally collected field data to validate generalisation. The RINDM module's dynamic N–P–K updates enable the ensemble to reconfigure its recommendation after rainfall events—a capability absent from all surveyed systems—making CropSense uniquely suited for within-season adaptive planning.

XII. Conclusion and Future Work

This paper presented **CropSense**, data-driven crop planning framework that integrates ensemble machine learning, physics-inspired soil nutrient dynamics, temporal forecasting, and reinforcement learning into a single continuously adaptive decision-support system.

The key contributions and results are:

1) A **soft-voting ensemble** (RF, XGBoost, CatBoost, SVM) achieving **99.55% test accuracy** across 22 crop classes.
2) **RINDM**, the first rainfall-induced nutrient depletion model in crop planning, providing dynamic N–P–K updates using leaching and runoff coefficients parameterised by soil texture and slope.
3) A **Temporal Meta Modeling** layer using LSTM for soil degradation, ARIMA for short-term rainfall, and Prophet for market price forecasting, enabling multi-season planning under uncertainty.
4) **Monte Carlo simulation** (10,000 scenarios) quantifying profit risk through VaR, CVaR, and loss probability, giving farmers actionable risk profiles.
5) **Q-Learning RL** optimising multi-season crop rotation policy that balances long-term profit with soil sustainability.

6) A **full-stack deployment** (Flask API + React/Vite + PostgreSQL) demonstrating production readiness.

Future work will focus on:

- Expanding to imbalanced, regionally sourced field datasets to validate real-world generalisation.
- Incorporating satellite-derived features (NDVI, soil moisture) for enhanced spatial awareness.
- Developing a mobile application for low-bandwidth rural environments.
- Integrating SHAP-based explainability to justify crop recommendations to farmers in their local language.
- Completing and evaluating the full Q-Learning deployment for 5-year rotation policy learning.
- Connecting IoT soil sensors for real-time N–P–K updates as input to RINDM.

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